Principal Component Analysis

ASM Presentation

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Multivariate Data Overview

- We consider a dataset where each observation consists of multiple measurements.
- Let $\mathbf{X} \in \mathbb{R}^p$ represent a vector of p variables or indicators.
- **Data Matrix:** Typically represented as an $n \times p$ matrix, where n is the number of observations and p is the number of features.
- The challenge lies in analyzing these p dimensions simultaneously to find hidden patterns.

Why Reduce Dimensionality?

- **Redundancy:** High-dimensional data often contains highly correlated variables, leading to redundant information.
- **Noise Reduction:** By keeping only the most significant components, we can filter out random noise.
- **Computational Efficiency:** Reducing the number of features speeds up subsequent statistical modeling.

The PCA Approach: Projection

- PCA seeks to project the original data onto a new direction (axis).
- **Linear Combination:** We define a new variable Y as a linear combination of the original features:
$$Y = \delta^T \mathbf{X} = \delta_1 X_1 + \delta_2 X_2 + \cdots + \delta_p X_p$$
- **The Optimization Goal:** We choose the direction δ such that we **maximize the variance** of Y.
- **Logic:** Higher variance means the projection captures the maximum amount of information (spread) from the original data.

The Goal of PCA

- **Main Objective:** Reduce the dimension of the observations.
- Naive Method would be to discard variables which can lead to loss of important information.
- Equal weighting ignores the relative importance of different variables.
- **Solution:** Create an optimal weighted average of the features.

Optimization Problem

- **Objective:** Maximize the variance of the projection

$$\boxed{\max_{\delta} \ \mathrm{Var}(\delta^T X)}$$

- **Constraint:** $\delta^T \delta = 1$
- **Assumption:** $\mathbb{E}[X] = 0$

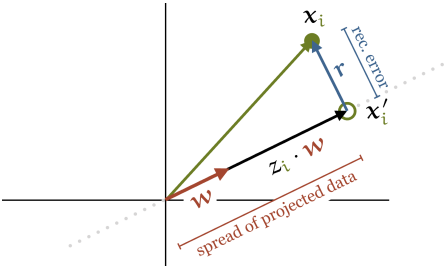


Figure: Minimizing orthogonal error maximizes variance.

- **Why Maximize Variance?**

$$\boxed{\text{Total Var}(X) = \text{Projected Var} + \text{Reconstruction Error} = \textbf{constant}}$$

Solving via Lagrange Multipliers

- First, express the variance using the covariance matrix $\Sigma = \text{Var}(X)$:

$$\text{Var}(\delta^T X) = \delta^T \Sigma \delta$$

- To maximize this subject to the unit-length constraint ($\delta^T \delta = 1$), construct the **Lagrangian** :

$$\mathcal{L}(\delta, \lambda) = \delta^T \Sigma \delta - \lambda(\delta^T \delta - 1)$$

- Differentiate with respect to the vector δ and set to zero to find the maximum:

$$\frac{\partial \mathcal{L}}{\partial \delta} = 2\Sigma\delta - 2\lambda\delta = 0$$

The First Principal Component Result

- Rearranging the derivative from the Lagrangian yields the fundamental mathematical solution for PCA:

$$\boxed{\Sigma\delta = \lambda\delta}$$

- **Key Result:** The direction that maximizes variance must be an **eigenvector** of the covariance matrix Σ.
- To capture the absolute maximum variance, we select the eigenvector γ_1 corresponding to the **largest eigenvalue** λ_1.

- **First Principal Component (PC1):**

$$Y_1 = \gamma_1^T (X - \mu)$$

- $\text{Var}(Y_1) = \lambda_1$ (The maximum possible variance for any linear combination).

Finding Subsequent Components

- After finding γ_1, we seek a second direction γ_2 that also maximizes variance but contains **new** information.
- **The Orthogonality Constraint:** To avoid redundancy, γ_2 must be perpendicular to γ_1 ($\gamma_2^T \gamma_1 = 0$).

The Iterative Optimization

- **2nd PC:** $\gamma_2 = \arg\max_{\|\delta\|=1, \delta \perp \gamma_1} \delta^T \Sigma \delta \implies$ 2nd largest eigenvalue.
- **k-th PC:** $\gamma_k = \arg\max_{\|\delta\|=1, \delta \perp \{\gamma_1, \ldots, \gamma_{k-1}\}} \delta^T \Sigma \delta \implies$ k-th largest eigenvalue.

- **Result:** We obtain an ordered set of p orthonormal eigenvectors $\{\gamma_1, \gamma_2, \ldots, \gamma_p\}$ representing decreasing levels of variance.

The Full PCA Transformation

- We stack these ordered eigenvectors to form an orthogonal transformation matrix Γ:

$$\boxed{\Gamma = [\gamma_1, \gamma_2, \ldots, \gamma_p]}$$

- **Coordinate Rotation:** We transform the centered data into the new "Principal Component" space:

$$\boxed{Y = \Gamma^T (X - \mu)}$$

- **Benefit:** This Y vector now represents the original data points in a new coordinate system where the axes are aligned with the directions of maximum spread.

Key Properties

- The new principal components are strictly uncorrelated (ensures no redundancy):

$$\boxed{\mathrm{Cov}(Y_i, Y_j) = 0 \quad \text{for } i \neq j}$$

- The variance of each component equals its corresponding eigenvalue:

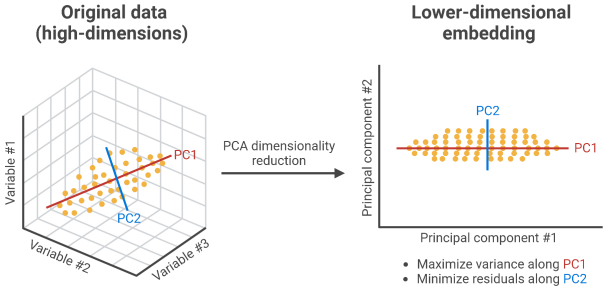
$$\boxed{\mathrm{Var}(Y_j) = \lambda_j}$$

- Components are naturally ordered by importance:

$$\boxed{\mathrm{Var}(Y_1) \geq \mathrm{Var}(Y_2) \geq \cdots \geq \mathrm{Var}(Y_p)}$$

Geometric View

- **Rotation:** PCA represents a rotation of the original coordinate system.
- **Preservation:** The intrinsic "shape" of the data remains the same, only the perspective changes.
- **Reduction:** Keep only the first P axes to perform an optimal lower-dimensional projection (These axes are exactly the eigenvectors we derived).



Summary

- **PCA:** Find linear combinations that maximize data variance.
- **Solution:** The eigenvectors of the covariance matrix Σ.
- **Eigenvalues:** A direct measure of importance (variance explained).
- **Output:** A rotated, decorrelated dataset strictly ordered by signal strength.

PCA is simply choosing the coordinate system where the data reveals itself most clearly.

Principal Component Analysis in Practice

- We now apply PCA to real data
- **Goal:** Reduce dimensionality while preserving variance

Iris Dataset

- 150 observations from **3 species of flowers**
- 4 features: Sepal Length, Sepal Width, Petal Length, Petal Width (in cm)
- Data matrix of size 150×4, **difficult to visualize in higher dimensions**

Note: The scale of all variables should be the same for PCA to give meaningful results.

Data Matrix and Sample Estimates

- Consider the data matrix:

$$\begin{array}{cccc} \text{(Sepal Length} & \text{Sepal Width} & \text{Petal Length} & \text{Petal Width)} \end{array}$$

$$\mathcal{X} = \begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{150\times 4}$$

- Replace population quantities:

$$\boldsymbol{\mu} \to \bar{\boldsymbol{x}}, \quad \boldsymbol{\Sigma} \to \mathcal{S}$$

- Sample mean:

$$\bar{\boldsymbol{x}}^\top = \begin{bmatrix} 5.843 & 3.057 & 3.758 & 1.199 \end{bmatrix}$$

Centering and Covariance

- **Centered data:**

$$\mathcal{X}_c = \mathcal{X} - 1_n \bar{\boldsymbol{x}}^\top$$

- **Sample covariance matrix:**

$$\mathcal{S} = \frac{1}{n-1} \mathcal{X}_c^\top \mathcal{X}_c$$

PCA Transformation

- **Eigen decomposition:**

$$\mathcal{S} = \mathcal{G}\mathcal{L}\mathcal{G}^\top$$

- **Empirical PC Transformation:**

$$\mathcal{Y} = (\mathcal{X} - 1_n \bar{\boldsymbol{x}}^\top)\mathcal{G}$$

Computation in R

Manual Computation

```
data(iris)
X <- as.matrix(iris[,1:4])
mu <- colMeans(X)
Xc <- sweep(X,2,mu)
S <- t(Xc) %*% Xc / nrow(X)
eig <- eigen(S)
Y <- Xc %*% eig$vectors
Y_2 <- Y[,1:2]
```

Built-in PCA

```
pca <- prcomp(X,
center = TRUE,
scale. = FALSE)

Y_2_prcomp <-
pca$x[,1:2]

Y_2_prcomp
```

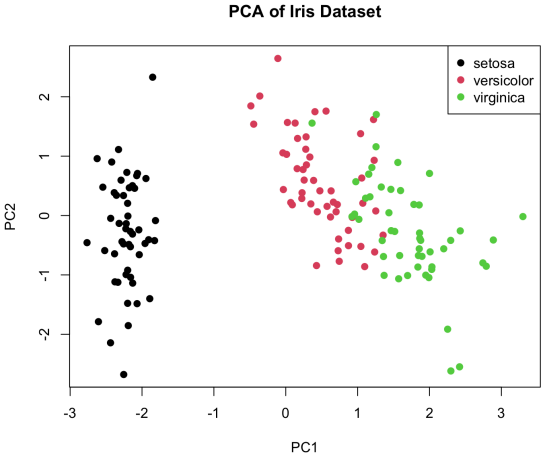
Transformed Matrix $\mathcal{Y}$

$$\mathcal{Y} = \begin{bmatrix} -2.68 & -0.32 & -0.03 & 0.00 \\ -2.71 & 0.18 & -0.21 & 0.10 \\ -2.89 & 0.14 & 0.02 & 0.02 \\ -2.75 & 0.32 & 0.03 & -0.08 \\ -2.73 & -0.33 & 0.09 & -0.06 \\ -2.28 & -0.74 & 0.17 & -0.02 \\ -2.82 & 0.09 & 0.26 & -0.05 \\ -2.63 & -0.16 & -0.02 & -0.05 \\ -2.89 & 0.58 & 0.02 & -0.03 \\ -2.67 & 0.11 & -0.20 & -0.06 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{150\times 4} \qquad \mathcal{Y}_{(2)} = \begin{bmatrix} -2.68 & -0.32 \\ -2.71 & 0.18 \\ -2.89 & 0.14 \\ -2.75 & 0.32 \\ -2.73 & -0.33 \\ -2.28 & -0.74 \\ -2.82 & 0.09 \\ -2.63 & -0.16 \\ -2.89 & 0.58 \\ -2.67 & 0.11 \\ \vdots & \vdots \end{bmatrix}_{150\times 2}$$

- **Reason:** First two principal components capture most of the variance

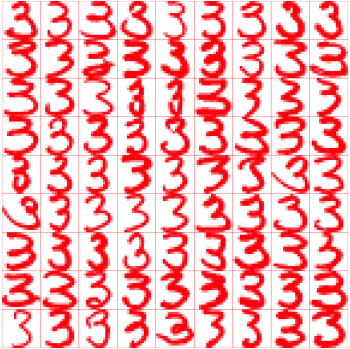
Visualization

- Reduced to **2 dimensions**
- **Pattern becomes visible**



Handwritten Threes Dataset

- Subset of handwritten digit images (ZIP dataset)
- Each image is of size 16×16 pixels
- Each image is represented as a vector in $\mathbb{R}^{256}$
- High-dimensional data with 256 features & 650 images of the digit 3



PCA Transformation (again!!)

- Center the data:

$$X_c = X - \mathbf{1}_n \bar{x}^T$$

- Covariance matrix:

$$S = \frac{1}{n-1} X_c^T X_c$$

- Spectral decomposition:

$$S = G \Lambda G^T$$

- PCA transformation:

$$Y = X_c G$$

Reconstruction

$$Y = (X - \mathbf{1}_n \bar{x}^T) G$$

$$YG^{-1} = X - \mathbf{1}_n \bar{x}^T$$

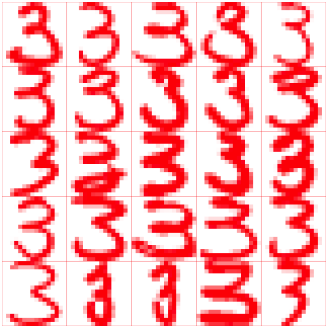
$$X = \mathbf{1}_n \bar{x}^T + YG^T$$

Final Reconstruction

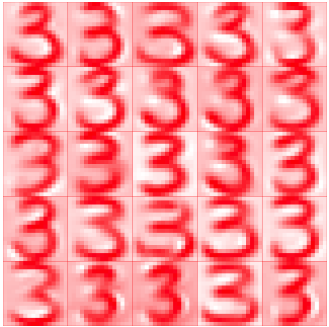
$$\hat{X}_{(q)} = \mathbf{1}_n \bar{x}^T + Y_{(q)} G_{(q)}^T$$

Reconstructed Images

Original



10 Principal Components



Using only 3.9% of the original features, we are able to reconstruct the image with reasonable accuracy.

Interpreting the Weights: Example

Based on our eigenvectors, we can construct the first two PCs:

$$y_1 = 0.361x_1 - 0.085x_2 + 0.857x_3 + 0.358x_4$$
$$y_2 = 0.657x_1 + 0.730x_2 - 0.173x_3 - 0.075x_4$$

Interpretation:

- y_1 represents the common variation of x_1, x_3, and x_4, with a slight opposite effect from x_2.
- y_2 primarily reflects the joint variation of x_1 and x_2, with negligible contribution from x_3 and x_4.

Proportion of Explained Variance

- A measure of how well the first q PCs explain variation is given by the relative proportion, ψ_q.
- Formula:

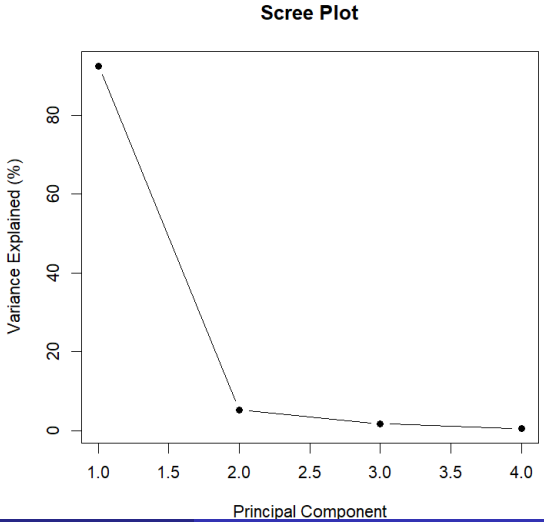
$$\psi_q = \frac{\sum_{j=1}^{q} \lambda_j}{\sum_{j=1}^{p} \lambda_j} = \frac{\sum_{j=1}^{q} \mathrm{Var}(Y_j)}{\sum_{j=1}^{p} \mathrm{Var}(Y_j)}$$

- This tells us the percentage of total variation captured if we reduce our data down to exactly q dimensions.

Visualizing Explained Variance: Scree Plot

PC	Var (%)	Cum. (%)
1	92%	92%
2	5%	97%
3	2%	99%

The first two PCs explain 97% of the total variation.



Correlation: Original Variables and PCs

To evaluate relations between the PCs (Y) and original variables (X), we calculate their covariance:

$$
\begin{aligned}
Cov(X, Y) &= E(XY^\top) - E(X)E(Y^\top) \\
&= E(XX^\top\Gamma) - \mu\mu^\top\Gamma \\
&= Var(X)\Gamma \\
&= \Sigma\Gamma = \Gamma\Lambda
\end{aligned}
$$

Sample Correlation Formula

Translating the population covariance to actual sample data, the correlation r between variable X_i and PC Y_j is:

$$r_{X_i Y_j} = g_{ij} \left(\frac{l_j}{s_{X_i X_i}} \right)^{1/2}$$

Where:

- g_{ij} is the eigenvector component
- l_j is the eigenvalue (variance of the PC)
- $s_{X_i X_i}$ is the sample variance of the original variable

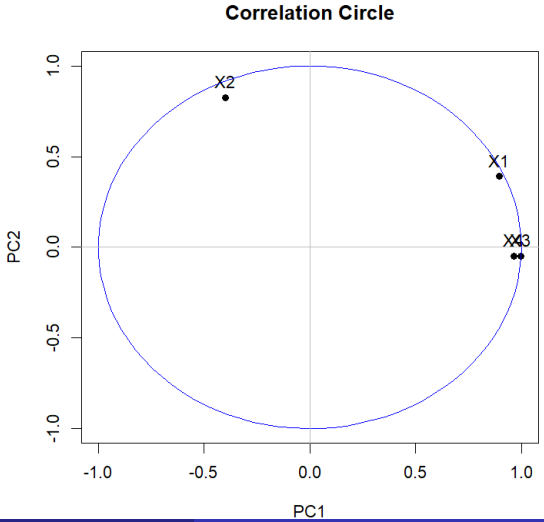
Variance Explained by a PC

- The squared correlation, $r^2_{X_i Y_j}$, represents the proportion of variance of the original variable X_i that is explained by Y_j.
- Because the PCs are orthogonal transformations of the original space, the sum of the squared correlations across all p principal components equals 1:

$$\sum_{j=1}^{p} r^2_{X_i Y_j} = \frac{\sum_{j=1}^{p} l_j g_{ij}^2}{s_{X_i X_i}} = 1$$

The Correlation Circle

- We plot $r_{X_iY_1}$ vs. $r_{X_iY_2}$.
- Since $r^2_{X_iY_1} + r^2_{X_iY_2} \leq 1$, all points fall inside a unit circle.
- Variables near the periphery are well-explained by the first two PCs.



Summary

- **Weights:** The components of the eigenvectors tell us the directions in which the best variance explanation is obtained.
- **Overall Variance:** ψ_q and the scree plot measure and visualize how much total variation the first q PCs capture.
- **Individual Variance:** $r^2_{X_i Y_j}$ tells us how much of an individual original variable's variance is explained by a specific PC.
- **Visualization:** The correlation circle identifies which original variables drive the principal components.

Thank You!